Training the First Layer Only in Deep Fully Connected Networks

Empirical evidence and a task-aligned feature-dynamics analysis

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Abstract

Deep networks are ordinarily optimized end to end, yet recent analyses of hierarchical teacher-student models suggest that depth can remain useful under highly restricted training. We study fully connected neural networks in which only the input layer is updated while all upper layers remain fixed at random initialization. Experiments cover a synthetic hierarchical target and binary CIFAR-10 tasks, comparing full training, extremities training, first-layer-only training, and first-layer training followed by temporary last-layer fine-tuning. On the synthetic task, a three-layer student trained only through its first layer reaches or exceeds the reported fully trained baselines. On CIFAR-10, width-aware experiments based on maximal-update parametrization show that first-layer-only models remain competitive across depths and widths, without establishing a universal advantage of depth or partial training. A first-order analysis shows that the induced feature update is a residual-weighted transport through the frozen upper layers and that prediction dynamics are governed by the neural tangent kernel restricted to first-layer parameters. This restricted kernel is positive semidefinite, providing local descent and, near small-output initialization, increasing an unnormalized target-alignment criterion. The results explain how task-aligned upper features can emerge even when those layers are frozen, while clarifying the limits of the claim.

Keywords— partial training; feature learning; multilayer perceptrons; neural tangent kernel; maximal-update parametrization; hierarchical targets.

1. Introduction

The usefulness of depth is usually discussed together with end-to-end optimization: each layer is expected to learn a representation that prepares the next layer for the task. Recent work on high-dimensional hierarchical targets gives this intuition a precise form. For a class of teacher functions with nested latent structure, gradient descent can progressively reduce the effective dimension of the problem, allowing a sufficiently deep student to learn with substantially fewer samples than a shallower one (Dandi and al. 2025).

This picture raises a sharper question. Is it necessary to train every layer for a deep architecture to exploit its composition, or can a single trainable input layer adapt its outputs to a fixed random nonlinear map above it? The latter regime is unusual. It is not a standard random-feature model, because the input representation changes; it is not ordinary feature learning either, because the downstream feature map is frozen. The trainable first layer must therefore learn *through* a fixed random computation.

The evidence provided in this paper supports a narrow but meaningful conclusion: a trainable first layer can exploit a fixed deep random map to produce task-aligned internal features. It does not yet show that first-layer-only training is generally better, cheaper end to end, or increasingly effective at arbitrary depth.

2. Background and research questions

2.1 Hierarchical targets and depth

Dandi and al. introduce high-dimensional hierarchical target functions in which successive nonlinear transformations reveal lower-dimensional latent structure (Dandi et al. 2025). Their analysis shows a computational advantage of depth in a controlled gradient-based training scheme. An experiment in that work also reports increasing alignment at an upper representation even when only the student's first layer is updated. This observation motivates two questions:

RQ1. Can a fully connected network learn a useful internal representation when only its first layer is trained?

RQ2. Can a network benefit from additional frozen depth under this training constraint?

These questions sit between classical random features, where a fixed nonlinear representation supports a trained linear predictor (Rahimi and Recht 2007), making learning easier. They are also naturally related to the neural tangent kernel (NTK), which describes first-order function-space dynamics under parameter gradient descent (Jacot, Gabriel, and Hongler 2018). Here the relevant object is not the full NTK but the contribution generated by the first-layer parameter block alone.

3. Experimental setting

3.1 Student architecture

For the theoretical analysis, the three-layer scalar-output multilayer perceptron is

$$\begin{aligned} \hat{f}(x) &= a^\top \phi(z_2(x)), \\ z_2(x) &= W_2 \phi(z_1(x)), \qquad z_1(x) = W_1 x. \end{aligned}$$

where $x \in \mathbb{R}^d$, $W_1 \in \mathbb{R}^{m_1 \times d}$, $W_2 \in \mathbb{R}^{m_2 \times m_1}$, and $a \in \mathbb{R}^{m_2}$. Biases are omitted for the sake of simplicity. The experiments use fully connected students with two, three, or four layers and hidden width $m \in \{512, 2048, 8192\}$.

3.2 Training regimes

Training regimes used in the experiments are described in **Table 1**. The last protocol decouples representation learning from readout fitting (last-layer fine-tuning). It should not be confused with simultaneous extreme layers training: last-layer optimization is performed on a temporary copy of the rest of the network at evaluation checkpoints.

3.3 Synthetic teacher-student task

Inputs are Gaussian, $x^\mu \sim \mathcal{N}(0, I_d)$, and labels are generated by the hierarchical teacher used in the motivating work (Dandi and al. 2025). In the reported experiment, $d = 100$, hierarchical index $k = 1$, sample size $n = 10^6$, hidden width $m = 512$, and student depth belongs to $\{2, 3, 4\}$. The task is binary classification derived from the teacher output.

Table 1. Optimization regimes. "First + last fine-tune" evaluates a temporary copy; it is not simultaneous extremities training.

Regime	First	Hidden	Output	Joint?	Evaluation procedure
Full	train	train	train	yes	Evaluate the jointly trained model.
Extremities	train	freeze	train	yes	Evaluate jointly trained first and output layers.
First only	train	freeze	freeze	no	Evaluate the partially trained model directly.
First + last FT	train	freeze	copy-only	no	Fine-tune the copied output layer, evaluate, discard the copy, and resume first-layer training.

3.4 Binary CIFAR-10 tasks

CIFAR-10 contains 60,000 color images in ten classes, with 50,000 training and 10,000 test images (Krizhevsky 2009). Because fully connected networks are poorly matched to image structure, the experiments restrict attention to binary class pairs. Two pairs are shown in the results: horse versus reindeer and cat versus dog. Widths $\{512, 2048, 8192\}$, depths $\{2,3,4\}$, and the four training regimes are compared.

To compare performances across width and depth without being biased by learning rate selection, we use μ-parametrization (muP) defined by Yang and al. (Yang and al. 2022). This rescaling method of the learning rate enables us to compare networks of different widths and depths with a unique learning rate. For the experiments, the method used is the following: (i) choosing one learning rate that performs similarly across widths and depths using muP, and (ii) repeating each reported configuration five times. The plotted intervals give the 95% confidence intervals for the estimated mean.

4. Empirical results

4.1 Synthetic task: frozen depth remains usable

The three- and four-layer students reach substantially higher accuracy than the two-layer student when only the first layer is trained. The first-layer-only models exceeds the displayed fully trained baselines. Extremities training and temporary last-layer fine-tuning are also competitive.

The loss trajectories provide a plausible regularization interpretation. Fully trained models drive training loss close to zero early, whereas first-layer-only models improve more gradually and retain a smaller displayed train–validation gap over much of training. This pattern is consistent with restricted optimization acting as an implicit capacity constraint. It is not, by itself, evidence that the full models are trapped in local minima: distinguishing optimization failure, overfitting, learning-rate mismatch, and early-stopping effects would require controlled ablations.

The synthetic results answer RQ1 positively within the studied setup: useful predictions emerge despite training only W_1. They also support RQ2 for this particular hierarchical teacher: additional fixed nonlinear depth is associated with a large improvement over the two-layer first-only model. Generalization beyond the chosen teacher, sample size, width, initialization, and optimizer remains open.

4.2 Binary CIFAR-10 under muP

Three conclusions are supported by **Figure 4**.

First, width matters: the 2,048- and 8,192-unit models generally outperform the 512-unit models. Second, once learning-rate scaling is controlled, fully trained models show no clear monotonic benefit from increasing depth over the tested range. Third, the partially trained regimes remain competitive. First-layer-only and extremities-trained models frequently lie near the best fully trained configurations, and temporary last-layer fine-tuning is competitive as well.

The differences are small relative to the plotted range, task dependent, and based on five runs. The result should therefore be framed as evidence of *viability*, not superiority. It answers RQ1 positively for these two binary tasks. RQ2 remains mixed on real data: fixed depth does not destroy performance and can sometimes help, but the plots do not establish a consistent depth advantage.

5. First-order theory of task-aligned feature evolution

We now show why the second-layer representation can move in a task-directed way even though W_2 is frozen.

Let $\ell_q = \ell(\hat{f}(x_q), y_q)$ and $r_q = \partial\ell(\hat{f}(x_q), y_q)/\partial\hat{f}$. For squared loss, $r_q = \hat{f}(x_q) - y_q$. Define

$$D_1(x) = \operatorname{diag}\big(\phi'(z_1(x))\big), \qquad D_2(x) = \operatorname{diag}\big(\phi'(z_2(x))\big),$$

and the back-propagated first-layer signal

$$g(x) = D_1(x) W_2^\top D_2(x) a \in \mathbb{R}^{m_1}.$$

The empirical objective is $L(W_1) = n^{-1}\sum_{q=1}^n \ell_q$, with W_2 and a fixed.

Proposition 1 (first-layer-induced feature update). For one full-batch gradient step $W_1^+ = W_1 - \eta\nabla_{W_1} L$, the second-layer preactivation at a test point x satisfies

$$z_2^+(x) - z_2(x) = -\frac{\eta}{n}\sum_{q=1}^n r_q\,(x^\top x_q)\,T(x, x_q) + O(\eta^2),$$

where the frozen-layer transport operator is

$$T(x, x_q) = W_2 D_1(x) D_1(x_q) W_2^\top D_2(x_q) a.$$

Proof. The chain rule gives

$$\nabla_{W_1}\hat{f}(x_q) = g(x_q)x_q^\top, \qquad \Delta W_1 = -\frac{\eta}{n}\sum_{q=1}^n r_q g(x_q) x_q^\top.$$

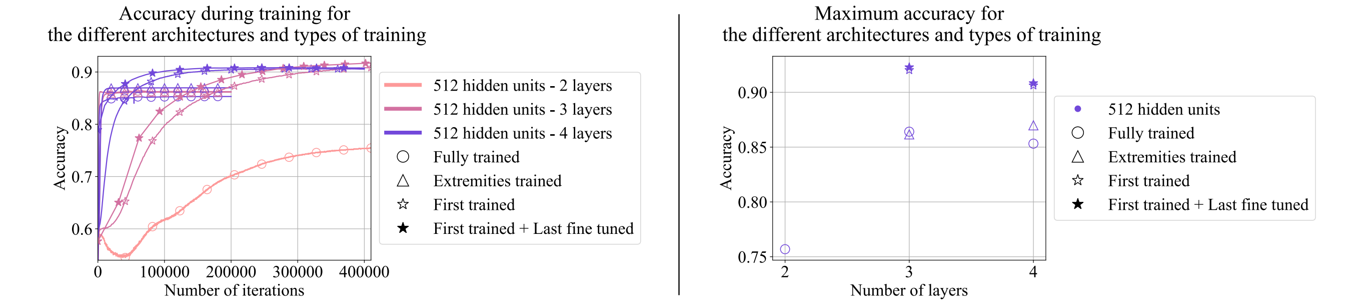

Figure 1. Synthetic teacher-student performance for width 512, $d = 100$, $k = 1$, and $n = 10^6$. Left: accuracy during training for the different architectures and training regimes. Right: maximum accuracy by depth and training regime.

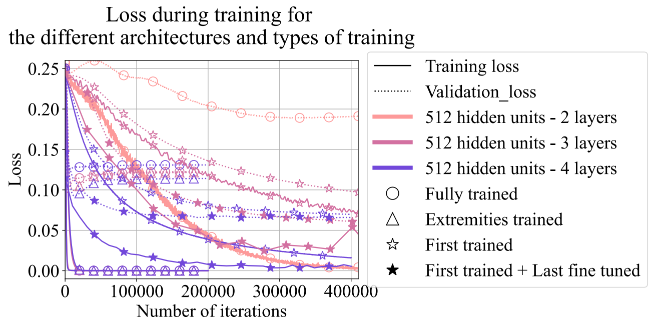

Figure 2. Training and validation mean-squared-error trajectories on the synthetic task. Solid and dotted curves denote training and validation loss, respectively.

Linearizing $z_2(x) = W_2\phi(W_1 x)$ around W_1 gives

$$\Delta z_2(x) = W_2 D_1(x)\Delta W_1 x + O(\|\Delta W_1\|^2).$$

Substitution of ΔW_1, followed by the definition of $g(x_q)$, yields the stated expression. $\square$

The update is a sum of training residuals transported through the fixed upper-layer Jacobians. Each example is weighted by the input inner product $x^\top x_q$ and by activation-gating overlap through $D_1(x)D_1(x_q)$. Thus, frozen W_2 does not imply frozen second-layer features: changing W_1 changes which first-layer neurons activate and how their outputs are mixed by W_2.

Corollary 1 (restricted-kernel prediction dynamics). To first order,

$$\hat{f}^+(x) - \hat{f}(x) = -\frac{\eta}{n}\sum_{q=1}^{n} K_1(x, x_q) r_q + O(\eta^2),$$

where

$$K_1(x, x') = \left\langle \nabla_{W_1}\hat{f}(x), \nabla_{W_1}\hat{f}(x') \right\rangle_F,$$

or, equivalently,

$$K_1(x, x') = (x^\top x')\, g(x)^\top g(x').$$

For any finite dataset, the Gram matrix $[K_1(x_p, x_q)]_{p,q}$ is positive semidefinite.

Proof. The first identity follows by linearizing $\hat{f}$ in W_1. Positive semidefiniteness follows because K_1 is an inner product between parameter gradients. Equivalently, it is the Gram matrix of the vectors $\mathrm{vec}(g(x_q)x_q^\top)$. $\square$

This is precisely the first-layer block of the empirical NTK (Jacot, Gabriel, and Hongler 2018). It proves that the prediction update is structured rather than random. More strongly, for sufficiently small η,

$$L(W_1^+) = L(W_1) - \eta\|\nabla_{W_1} L(W_1)\|_F^2 + O(\eta^2),$$

so training only W_1 is locally a descent method unless the restricted gradient vanishes.

Corollary 2 (initial target alignment for squared loss). Let $y = (y_1, \ldots, y_n)^\top$, $f = (\hat{f}(x_1), \ldots, \hat{f}(x_n))^\top$, and define the unnormalized alignment $A(f, y) = n^{-1} y^\top f$. If $f \approx 0$ at initialization, then

$$A(f^+, y) - A(f, y) = \frac{\eta}{n^2} y^\top K_1 y + O\big(\eta^2 + \eta\|f\|\big) \geq 0$$

to leading order.

The non-negativity follows from $K_1 \succeq 0$. This result formalizes the observation that small-output initialization can make an early first-layer step improve target alignment even though the upper layers are random and frozen.

5.1 What the analysis does and does not prove

The derivation supports three interpretations.

1. **Task dependence.** Labels enter through r_q, so feature motion is supervised rather than a generic drift of random representations.
2. **Neighborhood coupling.** Similarity is not only $x^\top x_q$; it also depends on shared activation gates and on how W_2 transports the back-propagated signal.
3. **Adaptive kernel.** Although only one parameter block is trained, K_1 generally changes with W_1. This is feature learning, not a fixed-kernel approximation, unless the network remains in a lazy regime.

The analysis does **not** imply that Pearson correlation with the target increases at every step. Away from the small-output regime,

$$y^\top \Delta f = -\frac{\eta}{n} y^\top K_1 (f - y) + O(\eta^2),$$

whose sign is not fixed. Nor does local loss descent guarantee superior validation accuracy. Those are empirical questions governed by kernel evolution, finite-step effects, optimization, and the train–test distribution.

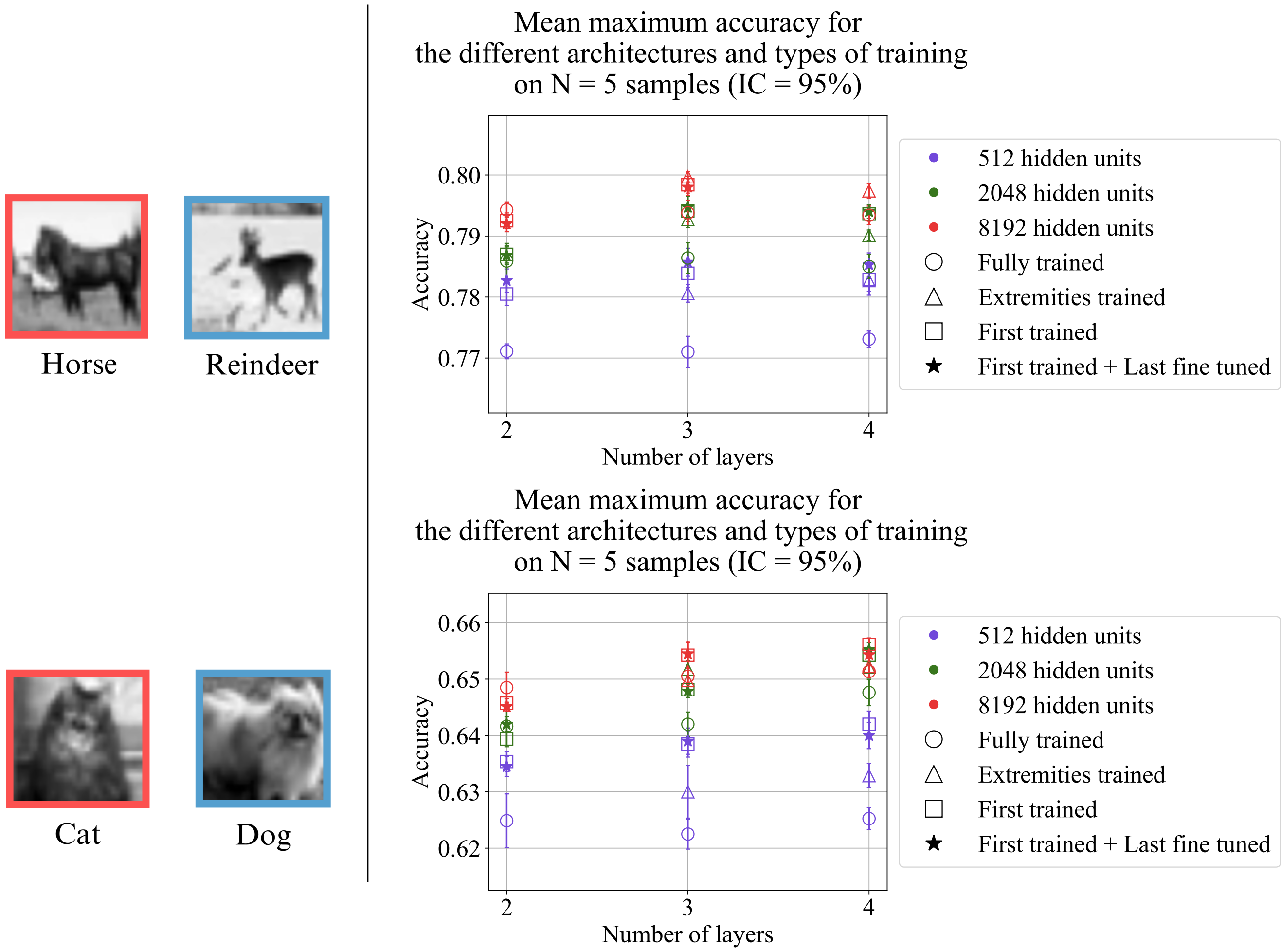

Figure 4. Mean maximum accuracy over five runs for binary CIFAR-10 tasks under muP. Error bars are reported as 95% confidence intervals in the source. Top: horse versus deer. Bottom: cat versus dog.

6. Discussion

6.1 How can a frozen deep map help?

The frozen upper network supplies a high-dimensional, nonlinear family of task gradients. Updating W_1 changes the input coordinates presented to this family. In effect, the first layer searches for a representation on which the fixed composition $a^\top \phi(W_2\phi(\cdot))$ becomes useful. Depth can enrich this induced function class even when its parameters are not trained, but it can also attenuate gradients or introduce unfavorable random geometry. The outcome therefore depends on initialization, width, nonlinearity, and target structure.

6.2 Partial training as regularization

Restricting trainable parameters reduces the accessible optimization directions. The synthetic loss curves are compatible with a regularization effect: slower fitting and a smaller displayed train–validation gap. However, parameter count alone does not determine effective capacity, and a frozen wide network can still define a rich function class. A convincing regularization claim would compare optimally early-stopped baselines, matched validation procedures, and several noise levels.

6.3 Efficiency claims require care

First-layer-only training reduces optimizer state and avoids storing gradients for frozen weights. It does not automatically remove the cost of the upper network: the forward pass is still required, and gradients must propagate through the frozen layers to reach W_1. Wall-clock time, activation memory, and energy use should therefore be measured directly before claiming end-to-end efficiency. The strongest present motivation is scientific—isolating where feature learning occurs—rather than an established systems-level speedup.

7. Conclusion

Training only the first layer of a deep fully connected network is sufficient to produce nontrivial learning in the studied synthetic and binary-image tasks. On the hierarchical teacher, frozen depth is associated with a pronounced improvement over a two-layer first-only model. On binary CIFAR-10, width-aware muP experiments show that partial-training regimes remain competitive, while weakening earlier claims of a general depth advantage for fully trained networks.

The theoretical mechanism is explicit. A first-layer gradient step moves second-layer preactivations through a residual-weighted combination of input similarity, activation-gate overlap, and fixed upper-layer Jacobians. At the output, this motion is governed by the positive-semidefinite first-layer block of the NTK. Consequently, the update is task aligned, locally decreases the objective, and near small-output initialization improves an unnormalized target-alignment criterion.

The next decisive step is to assess the generality of these findings under more demanding conditions. In particular, future work should evaluate first-layer-only training on substantially more challenging datasets and architectures, moving beyond fully connected networks to convolutional neural networks and other modern deep learning models. Such experiments would determine whether the representational advantages observed here persist in settings where hierarchical feature extraction is essential, or whether they are specific to the relatively simple tasks considered in this study. Establishing the robustness of these results across realistic benchmarks will be necessary to determine the practical scope of first-layer-only learning.

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Appendix - Detailed derivations for the first-order feature dynamics

This appendix provides the detailed calculations underlying Proposition 1 and Corollaries 1–2.

Network, notation, and regularity assumptions

Consider the scalar-output three-layer network

$$z_1(x) = W_1 x, \qquad h_1(x) = \phi(z_1(x)),$$
$$z_2(x) = W_2 h_1(x), \qquad h_2(x) = \phi(z_2(x)),$$
$$\hat{f}(x) = a^\top h_2(x).$$

The dimensions are

$$x \in \mathbb{R}^d, \qquad W_1 \in \mathbb{R}^{m_1 \times d}, \qquad W_2 \in \mathbb{R}^{m_2 \times m_1}, \qquad a \in \mathbb{R}^{m_2}.$$

Only W_1 is trained. The parameters W_2 and a remain fixed.
For each input x, define the diagonal activation-derivative matrices

$$D_1(x) = \operatorname{diag}\big(\phi'(z_1(x))\big) \in \mathbb{R}^{m_1 \times m_1},$$

and

$$D_2(x) = \operatorname{diag}\big(\phi'(z_2(x))\big) \in \mathbb{R}^{m_2 \times m_2}.$$

The back-propagated derivative of the output with respect to the first-layer preactivation is

$$g(x) = \nabla_{z_1(x)} \hat{f}(x) = D_1(x) W_2^\top D_2(x) a \in \mathbb{R}^{m_1}.$$

The empirical objective is

$$L(W_1) = \frac{1}{n} \sum_{q=1}^{n} \ell\big(\hat{f}(x_q), y_q\big).$$

For convenience, write

$$\ell_q = \ell\big(\hat{f}(x_q), y_q\big), \qquad r_q = \frac{\partial \ell(\hat{f}(x_q), y_q)}{\partial \hat{f}(x_q)}.$$

For the squared loss

$$\ell(\hat{f}, y) = \frac{1}{2}(\hat{f} - y)^2,$$

one has

$$r_q = \hat{f}(x_q) - y_q.$$

The derivations below assume that the network is locally twice differentiable with respect to W_1, so that the Taylor remainders are quadratic in the parameter displacement. For piecewise-linear activations such as ReLU, the same expressions hold locally as long as the gradient step does not cross an activation boundary. More generally, they hold almost everywhere and describe the first-order dynamics within a fixed activation region.

Gradient of the prediction with respect to the first layer

We first derive the identity

$$\nabla_{W_1} \hat{f}(x) = g(x) x^\top.$$

Let $\Delta W_1 \in \mathbb{R}^{m_1 \times d}$ be an arbitrary perturbation. The corresponding first-order change in the first-layer preactivation is

$$\Delta z_1(x) = \Delta W_1 x.$$

The first-order changes in the subsequent variables are

$$\Delta h_1(x) = D_1(x) \Delta z_1(x) = D_1(x) \Delta W_1 x,$$
$$\Delta z_2(x) = W_2 \Delta h_1(x) = W_2 D_1(x) \Delta W_1 x,$$

and

$$\Delta h_2(x) = D_2(x) \Delta z_2(x) = D_2(x) W_2 D_1(x) \Delta W_1 x.$$

Therefore,

$$\begin{aligned} \Delta \hat{f}(x) &= a^\top \Delta h_2(x) \\ &= a^\top D_2(x) W_2 D_1(x) \Delta W_1 x \\ &= g(x)^\top \Delta W_1 x. \end{aligned}$$

Using the Frobenius inner product

$$\langle A, B \rangle_F = \operatorname{Tr}(A^\top B),$$

we can rewrite this scalar as

$$g(x)^\top \Delta W_1 x = \operatorname{Tr}\left(x g(x)^\top \Delta W_1\right) = \left\langle g(x) x^\top, \Delta W_1 \right\rangle_F.$$

Since this holds for every perturbation ΔW_1,

$$\boxed{\nabla_{W_1} \hat{f}(x) = g(x) x^\top.}$$

This gradient is a rank-one matrix formed by the outer product of the back-propagated signal $g(x)$ and the input x.

Full-batch gradient step

Applying the chain rule to the empirical loss gives

$$\begin{aligned} \nabla_{W_1} L(W_1) &= \frac{1}{n} \sum_{q=1}^{n} r_q \nabla_{W_1} \hat{f}(x_q) \\ &= \frac{1}{n} \sum_{q=1}^{n} r_q g(x_q) x_q^\top. \end{aligned}$$

A full-batch gradient-descent step with learning rate η is therefore

$$W_1^+ = W_1 - \eta \nabla_{W_1} L(W_1),$$

or equivalently,

$$\boxed{\Delta W_1 := W_1^+ - W_1 = -\frac{\eta}{n} \sum_{q=1}^{n} r_q g(x_q) x_q^\top.}$$

Provided the gradients remain bounded locally,

$$\|\Delta W_1\|_F = O(\eta).$$

Consequently, a second-order Taylor remainder in ΔW_1 is $O(\eta^2)$.

Detailed proof of Proposition 1

At a test point x, the second-layer preactivation is

$$z_2(x) = W_2 \phi(W_1 x).$$

After the gradient step,

$$z_2^+(x) = W_2 \phi((W_1 + \Delta W_1) x).$$

A first-order Taylor expansion of ϕ around $W_1 x$ yields

$$\phi((W_1 + \Delta W_1) x) = \phi(W_1 x) + D_1(x) \Delta W_1 x + O(\|\Delta W_1 x\|^2).$$

Multiplying by W_2 gives

$$z_2^+(x) - z_2(x) = W_2 D_1(x) \Delta W_1 x + O(\|\Delta W_1\|_F^2).$$

Substituting the expression for ΔW_1,

$$\begin{aligned} z_2^+(x) - z_2(x) &= -\frac{\eta}{n} \sum_{q=1}^{n} r_q W_2 D_1(x) g(x_q) x_q^\top x + O(\eta^2) \\ &= -\frac{\eta}{n} \sum_{q=1}^{n} r_q (x^\top x_q) W_2 D_1(x) g(x_q) + O(\eta^2). \end{aligned}$$

Using

$$g(x_q) = D_1(x_q) W_2^\top D_2(x_q) a,$$

we obtain

$$W_2 D_1(x) g(x_q) = W_2 D_1(x) D_1(x_q) W_2^\top D_2(x_q) a.$$

Define

$$T(x, x_q) = W_2 D_1(x) D_1(x_q) W_2^\top D_2(x_q) a \in \mathbb{R}^{m_2}.$$

It follows that

$$\boxed{z_2^+(x) - z_2(x) = -\frac{\eta}{n} \sum_{q=1}^{n} r_q (x^\top x_q) T(x, x_q) + O(\eta^2).}$$

Each term in this sum contains four distinct components:

$$r_q, \qquad x^\top x_q, \qquad D_1(x) D_1(x_q), \qquad W_2^\top D_2(x_q) a.$$

The scalar r_q introduces task and label dependence. The input inner product $x^\top x_q$ couples geometrically similar examples. The product

$$D_1(x) D_1(x_q)$$

measures overlap between their active first-layer units. Finally, the fixed matrices W_2 and $W_2^\top$, together with the second-layer gate $D_2(x_q)$, transport the training signal through the frozen upper network.
Although W_2 is fixed, the representation $z_2(x)$ is not fixed, because it depends on the moving quantity

$$\phi(W_1 x).$$

Detailed derivation of Corollary 1

Taylor-expanding the scalar prediction at an arbitrary input x gives

$$\hat{f}^+(x) - \hat{f}(x) = \left\langle \nabla_{W_1} \hat{f}(x), \Delta W_1 \right\rangle_F + O(\|\Delta W_1\|_F^2).$$

Using

$$\nabla_{W_1} \hat{f}(x) = g(x) x^\top$$

and

$$\Delta W_1 = -\frac{\eta}{n} \sum_{q=1}^{n} r_q g(x_q) x_q^\top,$$

we obtain

$$\hat{f}^{+}(x)-\hat{f}(x)=-\frac{\eta}{n}\sum_{q=1}^{n}r_q\left\langle g(x)x^\top,g(x_q)x_q^\top\right\rangle_F+O(\eta^2).$$

Define the first-layer restricted kernel

$$K_1(x,x')=\langle\nabla_{W_1}\hat{f}(x),\nabla_{W_1}\hat{f}(x')\rangle_F\,.$$

Then

$$\boxed{\hat{f}^{+}(x)-\hat{f}(x)=-\frac{\eta}{n}\sum_{q=1}^{n}K_1(x,x_q)r_q+O(\eta^2).}$$

Factorized form of the kernel: The Frobenius inner product between two rank-one matrices satisfies

$$\left\langle uv^\top,st^\top\right\rangle_F=(u^\top s)(v^\top t).$$

Applying this identity with

$$u=g(x),\qquad v=x,\qquad s=g(x'),\qquad t=x',$$

gives

$$\begin{aligned}K_1(x,x')&=\left\langle g(x)x^\top,g(x')x'^\top\right\rangle_F\\&=\left(g(x)^\top g(x')\right)\left(x^\top x'\right).\end{aligned}$$

Hence

$$\boxed{K_1(x,x')=(x^\top x')g(x)^\top g(x').}$$

Equivalently, using the vectorization identity

$$\mathrm{vec}(uv^\top)=v\otimes u,$$

we may define the effective first-layer tangent feature

$$\psi_1(x)=\mathrm{vec}\left(\nabla_{W_1}\hat{f}(x)\right)=x\otimes g(x).$$

The kernel then takes the standard feature-inner-product form

$$K_1(x,x')=\psi_1(x)^\top\psi_1(x').$$

Thus the restricted tangent feature combines the original input x with the back-propagated feature $g(x)$.

Positive semidefiniteness: For a dataset

$$\mathscr{D}=\{x_1,\dots,x_n\},$$

let

$$\mathbf{K}_1=\left[K_1(x_p,x_q)\right]_{p,q=1}^{n}.$$

For any vector $c=(c_1,\dots,c_n)^\top\in\mathbb{R}^n$,

$$\begin{aligned}c^\top\mathbf{K}_1c&=\sum_{p=1}^{n}\sum_{q=1}^{n}c_pc_q\langle\nabla_{W_1}\hat{f}(x_p),\nabla_{W_1}\hat{f}(x_q)\rangle_F\\&=\left\langle\sum_{p=1}^{n}c_p\nabla_{W_1}\hat{f}(x_p),\sum_{q=1}^{n}c_q\nabla_{W_1}\hat{f}(x_q)\right\rangle_F\\&=\left\|\sum_{q=1}^{n}c_q\nabla_{W_1}\hat{f}(x_q)\right\|_F^2\\&\geq 0.\end{aligned}$$

Therefore,

$$\boxed{\mathbf{K}_1\succeq 0.}$$

Positive semidefiniteness does not require every entry $K_1(x_p,x_q)$ to be nonnegative. Individual off-diagonal entries may be negative. The relevant property is that every quadratic form $c^\top\mathbf{K}_1c$ is nonnegative.

Matrix form of the restricted-kernel dynamics

Define the vectors of training predictions and loss derivatives by

$$f=\begin{pmatrix}\hat{f}(x_1)\\ \vdots\\ \hat{f}(x_n)\end{pmatrix},\qquad r=\begin{pmatrix}r_1\\ \vdots\\ r_n\end{pmatrix}.$$

Applying the prediction update to all training inputs gives

$$\boxed{f^{+}-f=-\frac{\eta}{n}\mathbf{K}_1r+O(\eta^2).}$$

Here the $O(\eta^2)$ notation denotes a vector whose norm is bounded by a constant times η^2 locally.

For squared loss,

$$r=f-y,$$

so

$$f^{+}-f=-\frac{\eta}{n}\mathbf{K}_1(f-y)+O(\eta^2).$$

Since y is fixed,

$$r^{+}-r=f^{+}-f,$$

and therefore

$$\boxed{r^{+}=\left(I-\frac{\eta}{n}\mathbf{K}_1\right)r+O(\eta^2).}$$

At a fixed parameter value, let

$$\mathbf{K}_1=U\Lambda U^\top,$$

where

$$\Lambda=\mathrm{diag}(\lambda_1,\dots,\lambda_n),\qquad\lambda_i\geq 0.$$

If

$$r=\sum_{i=1}^{n}\alpha_iu_i,$$

then, to first order,

$$r^{+}=\sum_{i=1}^{n}\left(1-\frac{\eta}{n}\lambda_i\right)\alpha_iu_i+O(\eta^2).$$

Residual components aligned with eigenvectors having large eigenvalues are reduced more rapidly. Components in the null space of $\mathbf{K}_1$ are unchanged at first order.

This spectral interpretation applies instantaneously. Because $\mathbf{K}_1$ depends on W_1, both its eigenvalues and its eigenvectors may evolve during training.

Relation to the empirical NTK

Let θ denote the complete parameter vector of the network. The empirical neural tangent kernel is

$$K_{\mathrm{NTK}}(x,x')=\langle\nabla_\theta\hat{f}(x),\nabla_\theta\hat{f}(x')\rangle\,.$$

If the parameters are partitioned into blocks,

$$\theta=(W_1,W_2,a),$$

then

$$K_{\mathrm{NTK}}=K_{W_1}+K_{W_2}+K_a,$$

where each term is the inner product of the gradients with respect to the corresponding parameter block.

The kernel K_1 considered here is precisely

$$K_1=K_{W_1}.$$

Because only W_1 is optimized, the first-order prediction dynamics are governed by this restricted block rather than by the complete NTK.

Detailed loss-descent calculation

A Taylor expansion of the empirical objective around W_1 yields

$$L(W_1+\Delta W_1)=L(W_1)+\langle\nabla_{W_1}L(W_1),\Delta W_1\rangle_F+O(\|\Delta W_1\|_F^2).$$

For gradient descent,

$$\Delta W_1=-\eta\nabla_{W_1}L(W_1).$$

Consequently,

$$L(W_1^{+})=L(W_1)-\eta\left\|\nabla_{W_1}L(W_1)\right\|_F^2+O(\eta^2).$$

Thus

$$\boxed{L(W_1^{+})=L(W_1)-\eta\left\|\nabla_{W_1}L(W_1)\right\|_F^2+O(\eta^2).}$$

For sufficiently small $\eta>0$, the loss decreases whenever

$$\nabla_{W_1}L(W_1)\neq 0.$$

The same result can be written directly in terms of the restricted kernel. Since

$$\nabla_{W_1}L=\frac{1}{n}\sum_{q=1}^{n}r_q\nabla_{W_1}\hat{f}(x_q),$$

we have

$$\begin{aligned}\left\|\nabla_{W_1}L\right\|_F^2&=\frac{1}{n^2}\sum_{p=1}^{n}\sum_{q=1}^{n}r_pr_q\langle\nabla_{W_1}\hat{f}(x_p),\nabla_{W_1}\hat{f}(x_q)\rangle_F\\&=\frac{1}{n^2}r^\top\mathbf{K}_1r.\end{aligned}$$

Therefore,

$$\boxed{L(W_1^{+})-L(W_1)=-\frac{\eta}{n^2}r^\top\mathbf{K}_1r+O(\eta^2).}$$

Because $\mathbf{K}_1\succeq 0$,

$$r^\top\mathbf{K}_1r\geq 0.$$

Furthermore,
$$r^\top \mathbf{K}_1 r = 0$$
if and only if
$$\sum_{q=1}^{n} r_q \nabla_{W_1} \hat{f}(x_q) = 0,$$
which is equivalent to
$$\nabla_{W_1} L = 0.$$
The kernel expression therefore makes explicit that local descent occurs only through residual directions represented by the first-layer tangent features.

Detailed derivation of Corollary 2

For squared loss,
$$r = f - y.$$
The training-set prediction update is
$$f^+ - f = -\frac{\eta}{n}\mathbf{K}_1(f-y) + O(\eta^2).$$
Define the unnormalized target alignment
$$A(f,y) = \frac{1}{n} y^\top f.$$
Then
$$\begin{aligned} A(f^+,y) - A(f,y) &= \frac{1}{n} y^\top (f^+ - f) \\ &= -\frac{\eta}{n^2} y^\top \mathbf{K}_1 (f-y) + O(\eta^2) \\ &= \frac{\eta}{n^2} y^\top \mathbf{K}_1 y - \frac{\eta}{n^2} y^\top \mathbf{K}_1 f + O(\eta^2). \end{aligned}$$
Thus the exact first-order expression is
$$\boxed{A(f^+,y) - A(f,y) = \frac{\eta}{n^2} y^\top \mathbf{K}_1 y - \frac{\eta}{n^2} y^\top \mathbf{K}_1 f + O(\eta^2).}$$
Suppose the initialization produces small outputs,
$$\|f\| \ll \|y\|.$$
If the operator norm of $\mathbf{K}_1$ is locally bounded, then
$$\left| y^\top \mathbf{K}_1 f \right| \le \|y\| \|\mathbf{K}_1\|_{\mathrm{op}} \|f\|,$$
and hence
$$-\frac{\eta}{n^2} y^\top \mathbf{K}_1 f = O(\eta \|f\|).$$
It follows that
$$\boxed{A(f^+,y) - A(f,y) = \frac{\eta}{n^2} y^\top \mathbf{K}_1 y + O\big(\eta^2 + \eta \|f\|\big).}$$
Since
$$\mathbf{K}_1 \succeq 0,$$
the leading term satisfies
$$y^\top \mathbf{K}_1 y \ge 0.$$
Therefore, near a small-output initialization, the first gradient step cannot decrease the unnormalized alignment at leading order.
The leading-order gain is strictly positive if and only if
$$y \notin \ker(\mathbf{K}_1).$$
Equivalently, strict improvement requires the target vector to have a nonzero projection onto at least one tangent feature direction accessible through W_1.

Why this does not imply monotonic Pearson correlation

The quantity
$$A(f,y) = \frac{1}{n} y^\top f$$
is an unnormalized inner product. It differs from the Pearson correlation, which removes the sample means and normalizes by the standard deviations of f and y.
Even for the unnormalized alignment, away from the small-output regime,
$$y^\top (f^+ - f) = -\frac{\eta}{n} y^\top \mathbf{K}_1 (f-y) + O(\eta^2),$$
and the sign of
$$-y^\top \mathbf{K}_1 (f-y)$$
is not fixed.
Moreover, Pearson correlation depends not only on $y^\top f$, but also on the mean and norm of f. A gradient step can increase $y^\top f$ while changing these normalization terms in a way that decreases the correlation coefficient.
The alignment result therefore describes an initialization-dependent, leading-order tendency. It is not a global monotonicity theorem.

Adaptive rather than fixed-kernel dynamics

At iteration t, the restricted kernel is
$$K_1^{(t)}(x,x') = (x^\top x') g_t(x)^\top g_t(x'),$$
where
$$g_t(x) = D_{1,t}(x) W_2^\top D_{2,t}(x) a.$$
Although W_2 and a are fixed, both
$$D_{1,t}(x) \quad \text{and} \quad D_{2,t}(x)$$
can change as W_1 changes. Indeed,
$$z_{1,t}(x) = W_{1,t} x$$
changes directly, while
$$z_{2,t}(x) = W_2 \phi(W_{1,t} x)$$
changes indirectly.
Consequently,
$$K_1^{(t+1)} \neq K_1^{(t)}$$
in general.
The dynamics reduce to a fixed-kernel approximation only when the parameter displacement is sufficiently small that
$$g_t(x) \approx g_0(x)$$
throughout training. In that lazy regime,
$$K_1^{(t)}(x,x') \approx K_1^{(0)}(x,x').$$
Outside that regime, the evolving gates and back-propagated signals modify the tangent geometry. This evolution is the sense in which training only the first layer can still produce feature learning.

Continuous-time form

For completeness, consider gradient flow,
$$\frac{dW_1}{dt} = -\nabla_{W_1} L(W_1).$$
The prediction derivative at an arbitrary point x is
$$\begin{aligned} \frac{d\hat{f}(x)}{dt} &= \left\langle \nabla_{W_1} \hat{f}(x), \frac{dW_1}{dt} \right\rangle_F \\ &= -\frac{1}{n} \sum_{q=1}^{n} K_1(x,x_q) r_q. \end{aligned}$$
Thus
$$\boxed{\frac{d\hat{f}(x)}{dt} = -\frac{1}{n} \sum_{q=1}^{n} K_1(x,x_q) r_q.}$$
On the training set,
$$\boxed{\frac{df}{dt} = -\frac{1}{n}\mathbf{K}_1 r.}$$
For squared loss,
$$\frac{dr}{dt} = -\frac{1}{n}\mathbf{K}_1 r.$$
The loss satisfies
$$\begin{aligned} \frac{dL}{dt} &= \left\langle \nabla_{W_1} L, \frac{dW_1}{dt} \right\rangle_F \\ &= -\|\nabla_{W_1} L\|_F^2 \\ &= -\frac{1}{n^2} r^\top \mathbf{K}_1 r \le 0. \end{aligned}$$
Unlike the discrete-time statement, this loss-descent identity is exact along gradient flow rather than asymptotic in η.

Summary of the mechanism

The derivations establish the following sequence:
$$\text{training residuals} \longrightarrow \nabla_{W_1} L \longrightarrow \Delta W_1 \longrightarrow \Delta z_2 \longrightarrow \Delta \hat{f}.$$
More explicitly,
$$r_q$$
weights the contribution of training example q,
$$x^\top x_q$$
couples the test point to that example through input similarity,
$$D_1(x) D_1(x_q)$$

selects shared first-layer activation pathways, and

$$W_2^\top D_2(x_q)a$$

back-propagates the output-level task signal through the frozen upper network.

The resulting prediction dynamics are governed by the positive-semidefinite kernel

$$K_1(x,x') = (x^\top x')g(x)^\top g(x').$$

Thus, even with frozen upper-layer weights, the update is neither an arbitrary perturbation nor an unsupervised drift. It is a supervised gradient step constrained by the tangent feature space associated with the trainable first-layer parameter block.